DO NOW — RECALL

Database tables, entities and keys

Name: _____ Class: _____ Date: _____

1. Define INNER JOIN.

2. What is the purpose of a primary key?

3. Which SQL clause filters rows?

4. Insert 12 into BST with root 10 and right child 15. State its final position.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Combines matching table rows.

2. Model answer

Uniquely identifies each row.

3. Model answer

WHERE.

4. Model answer

12>10→right to15; 12<15→left child of15.

Vocabulary — know these terms

entity — Type of identifiable object.

attribute — Descriptive field of an entity.

primary key — Unique row identifier.

foreign key — Reference to key in another table.

index — Search aid on selected columns.

NEW LEARNING

Read, annotate and explain

Understand

An entity corresponds to a distinguishable object type; a relational table stores rows of its attributes.

Apply

A primary key uniquely identifies each row; foreign keys express references from related tables.

Evaluate

A secondary/index key can accelerate searching without necessarily being unique. One-to-many relationships use a foreign key on the many side.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Identify primary and foreign keys in Customer(CustomerID, Name) and Order(OrderID, CustomerID).

Predict first

Orders have their own identity and store a reference to the customer who placed them.

Model answer

Explain the reasoning and justify the output.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define entity, attribute, primary key, foreign key. [4]

Question 2

For Student, Book, Loan given above, state PKs and both FKs. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why storing every loan and every pupil detail in a single repeated row causes anomalies. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A foreign key must always be unique across every row in its table.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.